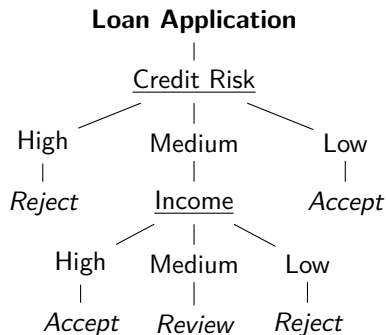
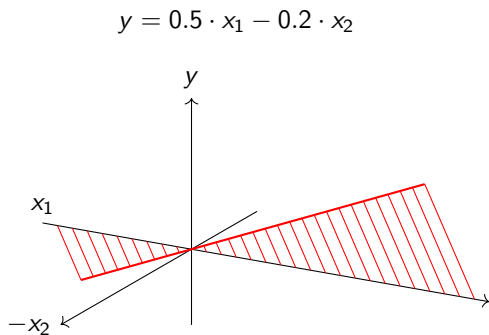


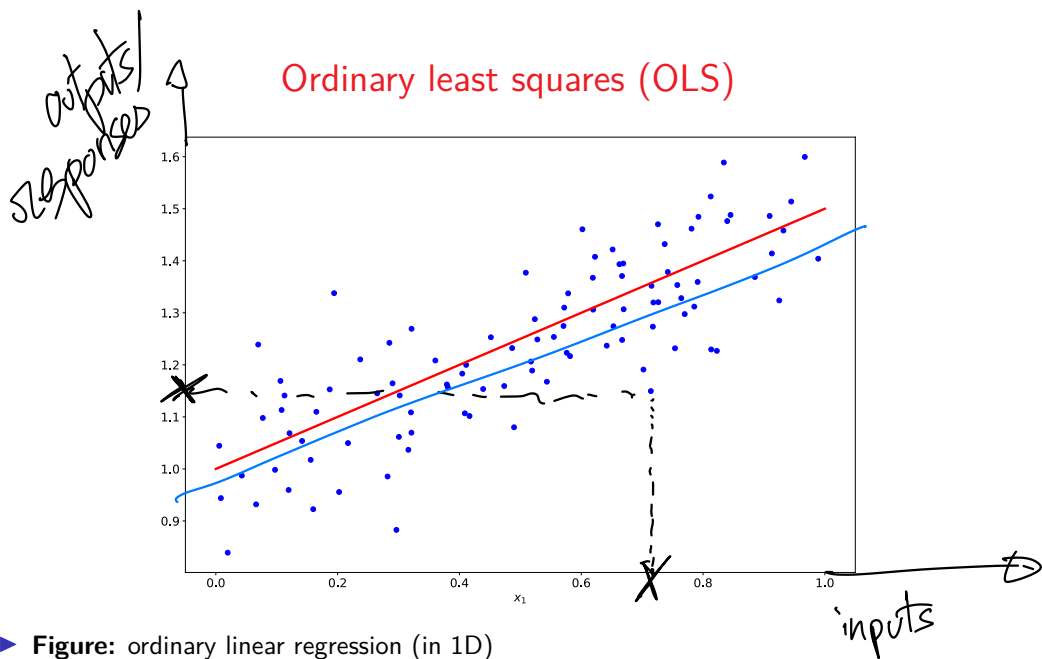
4. Interpretable-by-design models

Introduction

- ▶ for some models, **interpretability not an issue**
- ▶ **Examples:**
 - ▶ linear models (with few coefficients)
 - ▶ decision trees (with small depth and width)



4.1. Linear models



OLS: quick recap

features

- ▶ **Linear models:** output depends linearly on each feature
- ▶ mathematically, in the *regression* setting, our model is:

input

$$\forall x \in \mathbb{R}^d,$$

$$f(x) = \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_d x_d,$$

and we think of the data as $f(x) + \varepsilon$

- ▶ given training data $(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)})$, model is fitted by *ordinary least squares*

$$\hat{\beta} \in \arg \min_{\beta \in \mathbb{R}^d} \left\{ \sum_{i=1}^n \left(y^{(i)} - \sum_{j=1}^d \beta_j x_j^{(i)} \right)^2 \right\}$$

- ▶ **Intuition:** minimize the sum of squares of prediction errors

j-th feature
of i-th example.

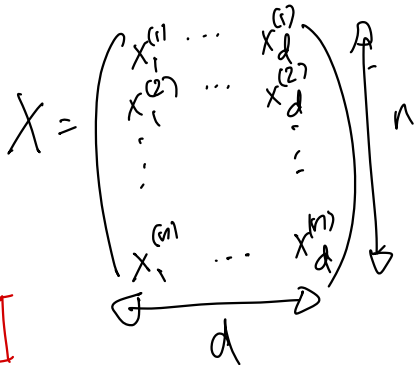
OLS: quick recap

- ▶ **Question:** how to do it, concretely?
- ▶ collect all inputs in *data matrix* $X \in \mathbb{R}^{n \times d}$
- ▶ collect all responses in $Y \in \mathbb{R}^n$
- ▶ **Claim:** if $X^\top X \in \mathbb{R}^{d \times d}$ is invertible, $\hat{\beta}$ is given by

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

- ▶ **Remark:** if $X^\top X$ is *not* invertible, take the Moore-Penrose pseudo-inverse
- ▶ generally a good idea to use existing libraries:

```
import numpy as np
from sklearn.linear_model import LinearRegression
X = some_data
y = some_responses
reg = LinearRegression().fit(X, y)
```



$$X^\top = \begin{pmatrix} x_1^{(1)} & \dots & x_1^{(n)} \\ x_d^{(1)} & \dots & x_d^{(n)} \end{pmatrix} \in \mathbb{R}^{d \times n}.$$

Controlling complexity

- ▶ usual ways to control the size / number of non-zeros coefficients: ridge¹⁶ and LASSO¹⁷
- ▶ **Ridge regression:** adding a L^2 penalty in the optimization:

finds good coef.
with small values.

$$\min_{\beta} \left\{ \sum_{i=1}^n \left(y^{(i)} - \sum_{j=1}^d \beta_j x_j^{(i)} \right)^2 + \lambda \sum_{j=1}^d \beta_j^2 \right\},$$

where $\lambda > 0$ is a hyperparameter

- ▶ **Least Absolute Shrinkage and Selection Operator (LASSO):** adding a L^1 penalty:

finds coef. with
lots of zeroes.

$$\min_{\beta} \left\{ \sum_{i=1}^n \left(y^{(i)} - \sum_{j=1}^d \beta_j x_j^{(i)} \right)^2 + \lambda \sum_{j=1}^d |\beta_j| \right\},$$

where $\lambda > 0$ is a hyperparameter

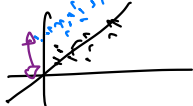
penalty

Same criterion
as OLS
= fit to the
data.

$$\beta_1 x_1 + \cancel{\beta_2 x_2} + \beta_3 x_3$$

¹⁶Hoerl and Kennard, *Ridge regression: Biased estimation for nonorthogonal problems*, Technometrics, 1970

¹⁷Tibshirani, *Regression shrinkage and selection via the lasso*, Journal of the Royal Statistical Society Series B: Statistical Methodology, 1996



Linear models: interpretability

$$f(x) = \dots + \beta_j x_j + \beta_0$$

$$\beta_j (x_j + 1) = \beta_j x_j + \beta_j$$

- ▶ **Why are these models interpretable?** increasing x_j by one unit increases $f(x)$ by β_j
- ▶ let us look at a concrete example: regression task on the California housing dataset¹⁸
- ▶ we run LASSO on a train set (75% of the data)
- ▶ RMSE on the test is 0.83 (not too bad!)
- ▶ we can read the coefficients:

$\beta_0 =$ `intercept: 0.285`
`coefficients: [ 0.346 0.015 -0. 0. 0. -0.001 -0. -0. ]`

β_1 β_2 β_8

- ▶ **Mathematically:** our model is given by

$$f(x) = 0.285 + 0.346 \cdot x_{\text{MedInc}} + 0.015 \cdot x_{\text{HouseAge}} - 0.001 \cdot x_{\text{AveOccup}}$$

- ▶ we can directly read in these coefficients what our model is doing!

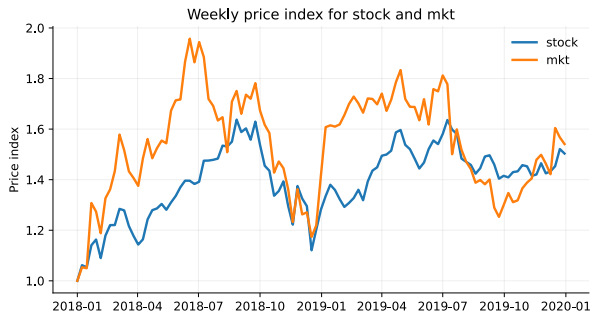
Intercept interpretation: depends on the normalization of the data

¹⁸Pace and Barry, *Sparse Spatial Autoregressions*, Statistics & Probability Letters, 1997

4.2. Vector autoregression (VAR)

Autoregressive forecast

- **Variation on this theme:** time series prediction
- **Example:** stock prices



- **Figure:** (normalized) stock prices for Amazon and the Nasdaq

Lags

$t = \text{time}$.

- ▶ **Question:** what are the features in that case?
- ▶ simply take the past values (= lags) = x_{t-s}
- ▶ **Formally:** predict $x_t \in \mathbb{R}$ using $x_{t-1}, \dots, x_{t-d}$ as

$$x_t = \sum_{j=1}^d a_j x_{t-j}, = a_1 x_{t-1} + a_2 x_{t-2} + \dots + a_d x_{t-d}.$$

where $a_1, \dots, a_d$ are the parameters of our model

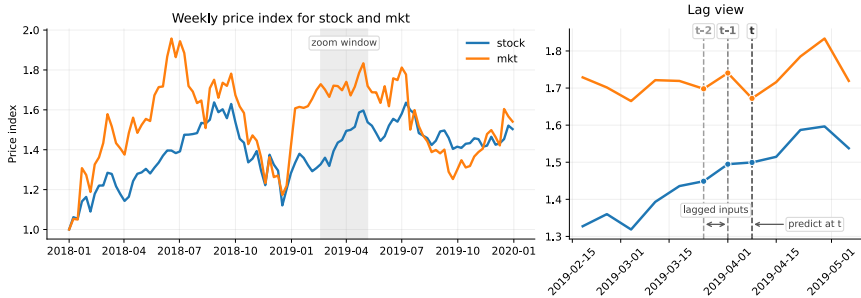
- ▶ we think of the data as $y_t = \sum_{j=1}^d a_j y_{t-j} + \varepsilon_t$, where ε_t is some noise
- ▶ to find the a_j s, two main options:
 - ▶ **least-squares;**
 - ▶ solving the **Yule-Walker**^{19,20} equations

$(x_1, x_2, x_3, \dots, x_d)$	x_{d+1}
$(x_2, x_3, \dots, x_{d+1})$	x_{d+2}
	$\vdots$

¹⁹Yule, *On a method of investigating periodicities disturbed series, with special reference to Wolfer's sunspot numbers*, Trans. Royal Soc., 1927

²⁰Walker, *On periodicity in series of related terms*, Phil. Trans. Royal Soc., 1931

Lags, illustrated



► **Figure:** lagged inputs for financial time series

Vector autoregression

- ▶ **Natural extension:** joint forecasting of time series
- ▶ to put it simply: $x_t \in \mathbb{R}^p$ = *p time series*
- ▶ the a_j s coefficients are replaced by a matrix
- ▶ **Example:** $d = 1$, $p = 2$, stock return and market return:

$$\begin{pmatrix} r_t^{\text{stock}} \\ r_t^{\text{mkt}} \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + A \begin{pmatrix} r_{t-1}^{\text{stock}} \\ r_{t-1}^{\text{mkt}} \end{pmatrix} + \begin{pmatrix} u_{1t} \\ u_{2t} \end{pmatrix}.$$

$$A \in \mathbb{R}^{2 \times 2}$$

- ▶ finding good parameters is similar (multivariate regression)
- ▶ we use the statsmodels package

$$\hat{A} \in \arg \min \dots$$

Vector autoregression

fitted model

- the fitted coefficients are

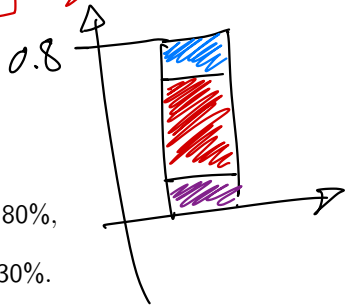
(*)
$$\begin{pmatrix} r_t^{\text{stock}} \\ r_t^{\text{mkt}} \end{pmatrix} = \begin{pmatrix} 0.10 \\ 0.05 \end{pmatrix} + \begin{pmatrix} 0.60 & 0.20 \\ -0.10 & 0.70 \end{pmatrix} \begin{pmatrix} r_{t-1}^{\text{stock}} \\ r_{t-1}^{\text{mkt}} \end{pmatrix} + \begin{pmatrix} u_{1t} \\ u_{2t} \end{pmatrix}$$

- assume last week's returns were

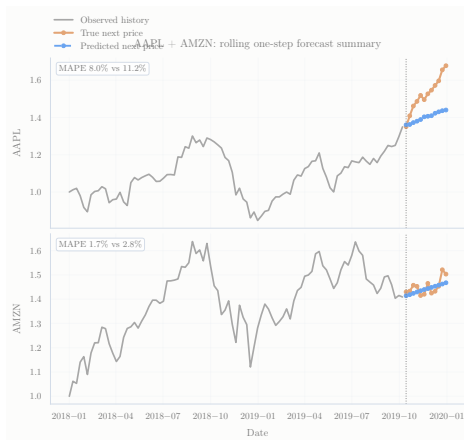
$$r_{t-1}^{\text{stock}} = 1.0\%, \quad r_{t-1}^{\text{mkt}} = 0.5\%.$$

- then the one-step-ahead forecasts are

(*)
$$\begin{aligned} \hat{r}_t^{\text{stock}} &= 0.10 + 0.60(1.0) + 0.20(0.5) = 0.80\%, \\ \hat{r}_t^{\text{mkt}} &= 0.05 - 0.10(1.0) + 0.70(0.5) = 0.30\%. \end{aligned}$$



Vector autoregression



- **Figure:** VAR in action: jointly predicting Amazon and Apple; MAPE = mean absolute percentage error

4.3. Generalized additive models (GAMs)

Generalized additive models (GAMs)

$$f(x) = \pi_1 \pi_2$$

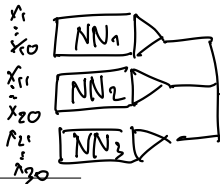
- **Straightforward extension of linear models:**²¹

$$\forall x \in \mathbb{R}^d, \quad f(x) = f_1(x_1) + f_2(x_2) + \dots + f_d(x_d),$$

where $f_1, \dots, f_d : \mathbb{R} \rightarrow \mathbb{R}$

- **Remark:** still no interactions ($\Rightarrow$ restrictive) $f_j(x_j) = \theta_j x_j$ (linear case) $\leftarrow$
- **Question:** what to choose for $f_1, \dots, f_d$?
- **Typical choices include:**

- polynomials
- splines
- neural nets²²
- ...

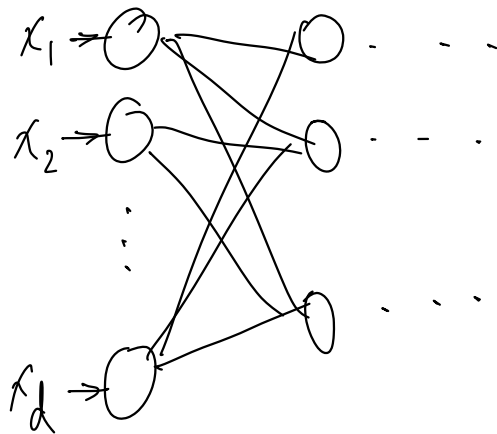


$$f(x) = NN_1(x) + NN_2(x) + NN_3(x)$$

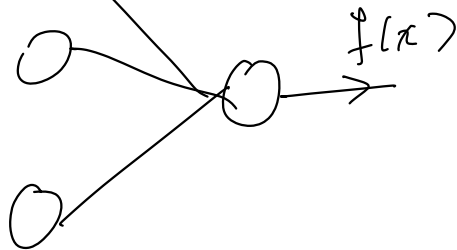
$\rightarrow$ find the paper.

²¹Hastie and Tibshirani, *Generalized Additive Models*, Monographs on Statistics and Probability, 1990

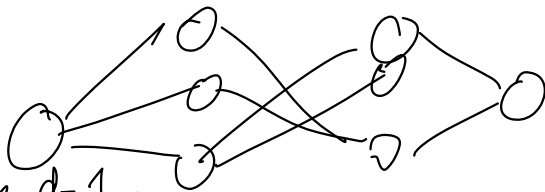
²²Agarwal et al., *Neural Additive Models: Interpretable Machine Learning with Neural Nets*, NeurIPS, 2021



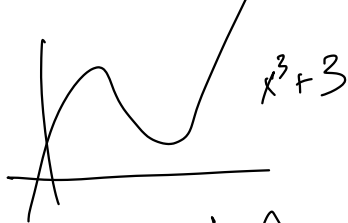
Problem: interactions



Remark: NN net so
interesting for $d=1$.



$$f(x) = f_1(x_1) + \dots + f_d(x_d) \quad \text{Splines}$$



- **Definition:** piecewise polynomial
- generally degree fixed by the user, complexity controlled by penalization
- **Typical objective:**

$$\sum_{t=1}^T (y_t - \hat{y}_t)^2 + \sum_{j=1}^d \lambda_j \int [f_j''(x)]^2 dx.$$

- $\lambda_1, \dots, \lambda_d$ control smoothness: large λ pushes towards straight line
- **Example:** continuing previous example:

$$r_t^{\text{stock}} = \beta_0 + f_1(r_{t-1}^{\text{stock}}) + f_2(r_{t-1}^{\text{mkt}}) + \varepsilon_t.$$

will punish



Q: Can we optimize the intervals?

Explaining GAMs

- ▶ **Two types of explanations:**

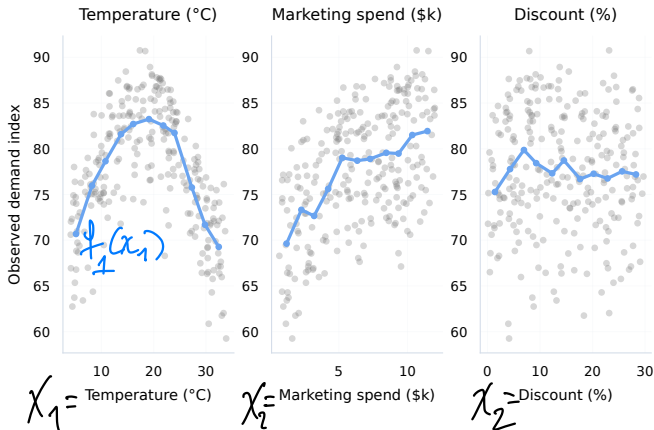
- ▶ decompose a given forecast (local explanation)
- ▶ plot f_j over data range (blue curve in previous slide)

- ▶ **Back to the example:** assume $r_{t-1}^{\text{stock}} = 1.0$ and $r_{t-1}^{\text{mkt}} = 0.5$,

$$r_t^{\text{stock}} = \hat{\beta}_0 + \hat{f}_1(1.0) + \hat{f}_2(0.5).$$

$$\implies 0.70 = 0.10 + 0.45 + 0.15.$$

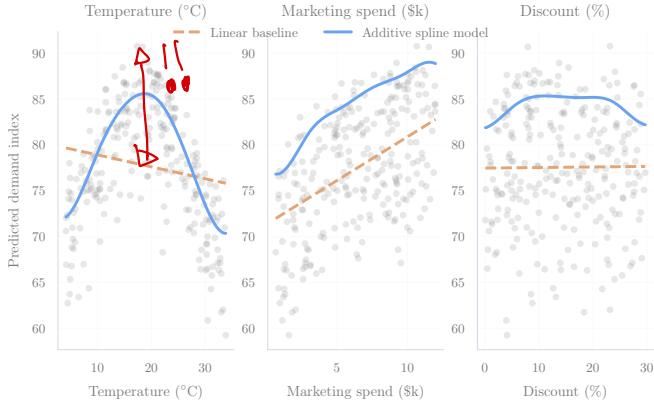
GAMs in action



► **Figure:** predicting consumer demand from three features

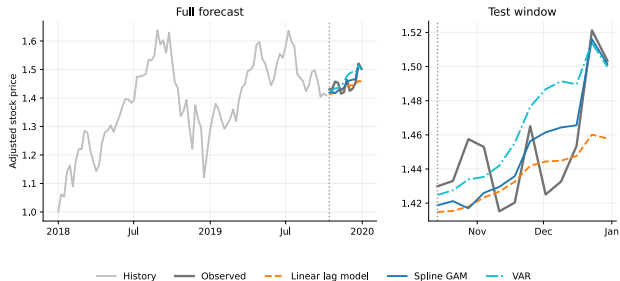
$$y = f(x) = f_1(x_1) + f_2(x_2) + f_3(x_3)$$

Comparison with linear model



► **Figure:** comparing with a linear model

Overall comparison



► **Figure:** comparing our favorite forecasters

Identifiability

- ▶ GAMs are not *identifiable*²³

- ▶ **Example:** say $y = f_1(x_1) + f_2(x_2) \approx f(x)$

- ▶ then

$$y = g_1(x_1) + g_2(x_2) = \underbrace{(f_1(x_1) - \Delta)}_{g_1} + \underbrace{(f_2(x_2) + \Delta)}_{g_2}$$

is an **equally valid decomposition**

- ▶ in this lecture, **we do not care, we want to explain the model!**
- ▶ but, if you want to explain the data, this is an issue
- ▶ **Partial solution:** force

$$\forall j \in [d], \quad \int f_j(x_j) d\mu(x_j) = 0.$$

²³Zhang et al., *Challenges in interpretability of additive models*, preprint, 2025

4.5. Decision trees

Decision tree: quick recap

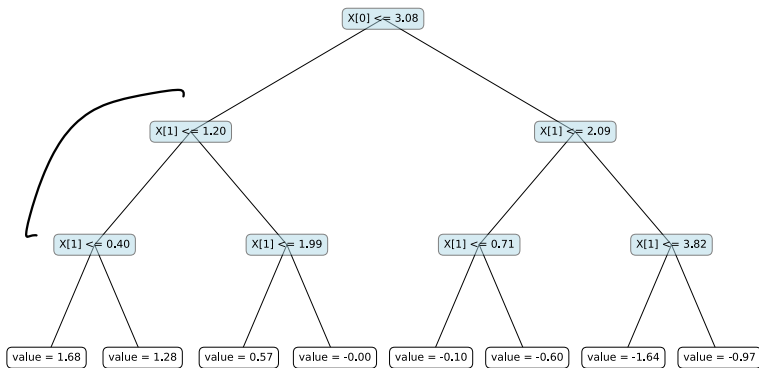
- ▶ **What is a decision tree?**

- ▶ tree with root = $\mathcal{X}$ and leaves = cells
- ▶ iterative binary decisions based on feature values
- ▶ node of the tree: "is feature j smaller than x ?"
- ▶ if yes, go left, if not, go right
- ▶ **Can also be visualized as** partition of the input space $\mathcal{X}$
- ▶ each query point falls into a cell, constant prediction on each cell

- ▶ two different modes:

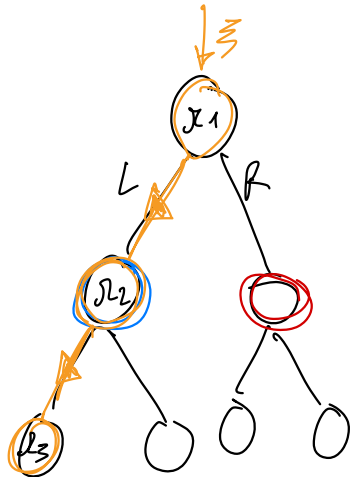
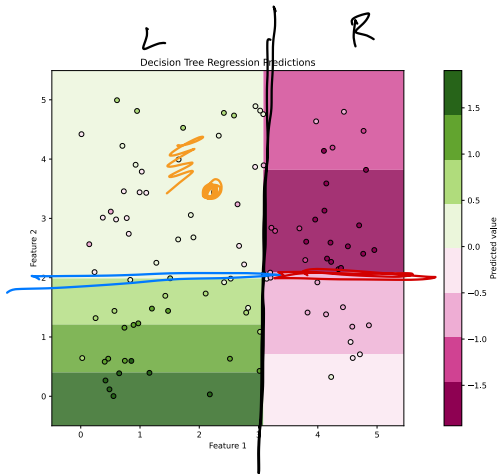
- ▶ classification → class label → **majority vote**
- ▶ regression → real number → **empirical average**

Decision tree: example



► **Figure:** example of a decision tree for regression in 2D

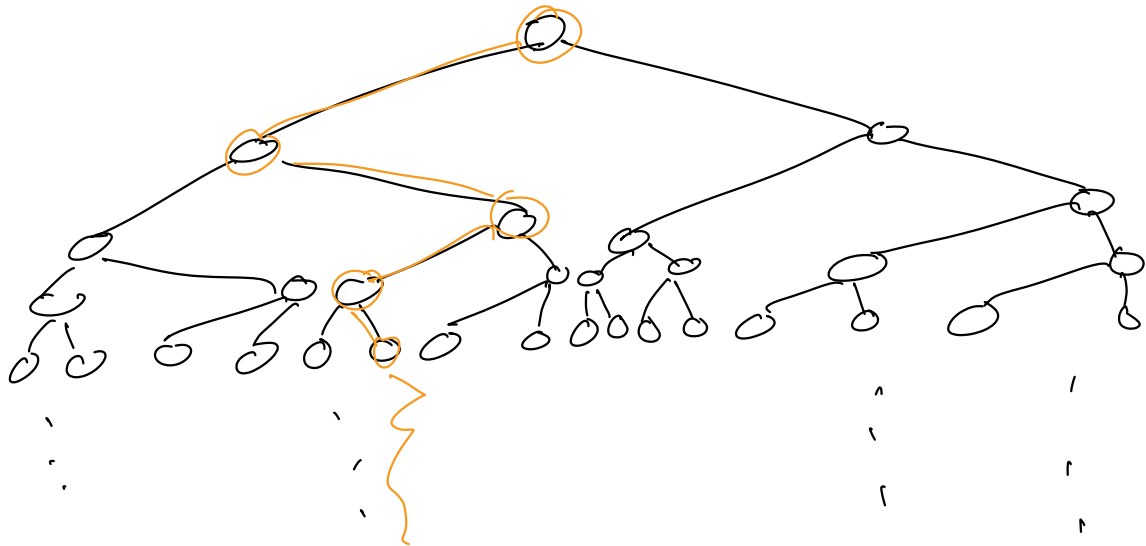
Decision tree: example



— = trace of $\mathbf{x}$.

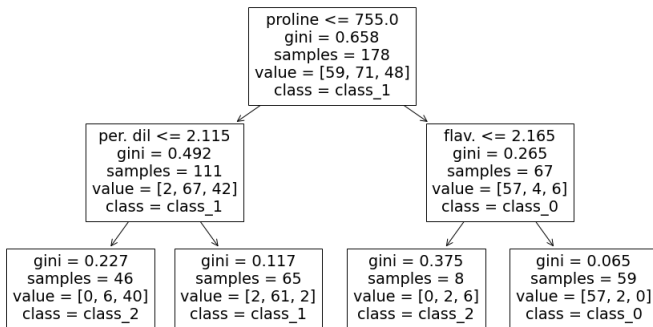
► **Figure:** associated partition of the space

" $\mathbf{x}$ got this value because x_1 and x_2 and x_3 ".



Decision trees: interpretability

- Why is this model interpretable?
- let us look at a concrete example: the Wine dataset²⁴

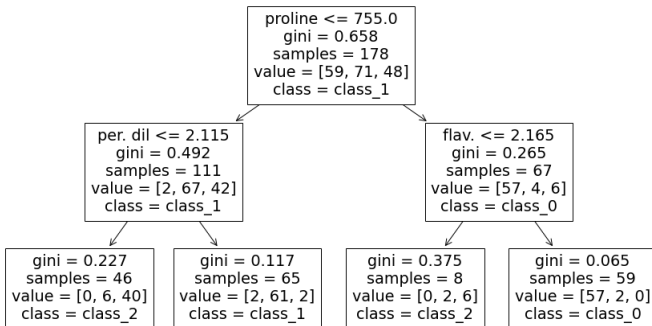


²⁴Cortez et al., *Modeling wine preferences by data mining from physicochemical properties*, Decision Support Systems, 1998

Decision trees: interpretability

- ▶ for a specific example, we can **run down the path** to understand the decision
- ▶ we can also infer **global rules**
- ▶ for instance, we know that

$\{\text{proline} \leq 755.0, \text{per. dil} \leq 2.115\} \rightarrow \text{class 2.}$



Summary

- ▶ decision trees are light-weight models
- ▶ recursively splitting the input space according to a numerical criterion
- ▶ **Limitations:**
 - ▶ accuracy far from state-of-the-art (model too simple)
 - ▶ interpretability decreases with number of leaves (see next section)
- ▶ **They are interpretable:** either tracing down the decision or deducing global rules
- ▶ possible to be more quantitative (mean decrease impurity / accuracy)²⁵
- ▶ extension to random forests possible

²⁵Breiman, *Random Forests*, Machine Learning, 2001